

Questions

- 1 Zara's mother opens a bottle of perfume. Zara is at the other side of the room.

After a few minutes, Zara can smell the perfume.

- a Which of these explains why Zara smells the perfume?
Write **one** letter.
- A All the gas particles of the perfume move in one direction.
 - B All the gas particles of the perfume move randomly.
 - C All the particles in the air move in one direction.
 - D All the particles in the air are stationary (not moving).
- b Which of these changes would result in Zara smelling the perfume in a shorter time?

In each case, **no** other variable changes.

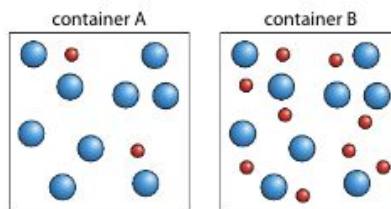
Write the letters for **all** the correct statements.

- A The perfume in the bottle is at a higher temperature.
- B Zara moves further away from the bottle of perfume.
- C Zara's mother puts the lid on the perfume bottle.
- D The air in the room is at a lower temperature.

- 2 Describe what is meant by the term 'diffusion'.

- 3 The diagram shows two types of particle in containers, A and B. The containers have the same volume.

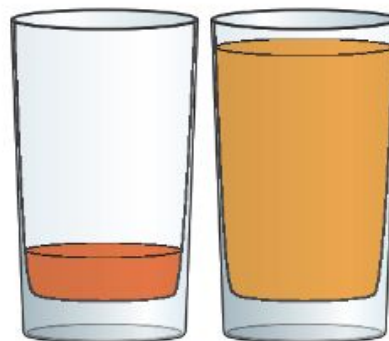
- a Explain whether the particles in the containers are in a solid, liquid or gas state.
- b Explain how the concentrations of the blue particles compare in containers A and B.
- c Explain how the concentrations of the red particles compare in containers A and B.



- 4 Sofia is making an orange flavour drink. She pours a small volume of concentrated orange juice into a glass. She then adds water until the glass is full.

- a Explain how the orange colour from the juice spreads into the water.
- b Sofia says: 'When the orange colour has stopped spreading, the particles in the liquid have stopped moving.'

Explain whether Sofia is correct.



- 5 Which of these will result in diffusion?

Write the letters for **all** the correct statements.

- A Adding milk to coffee.
- B Adding sand to water.
- C Adding salty water to pure water.
- D Allowing gas from a cylinder to escape into the air.
- E Throwing small pieces of rock into the air.

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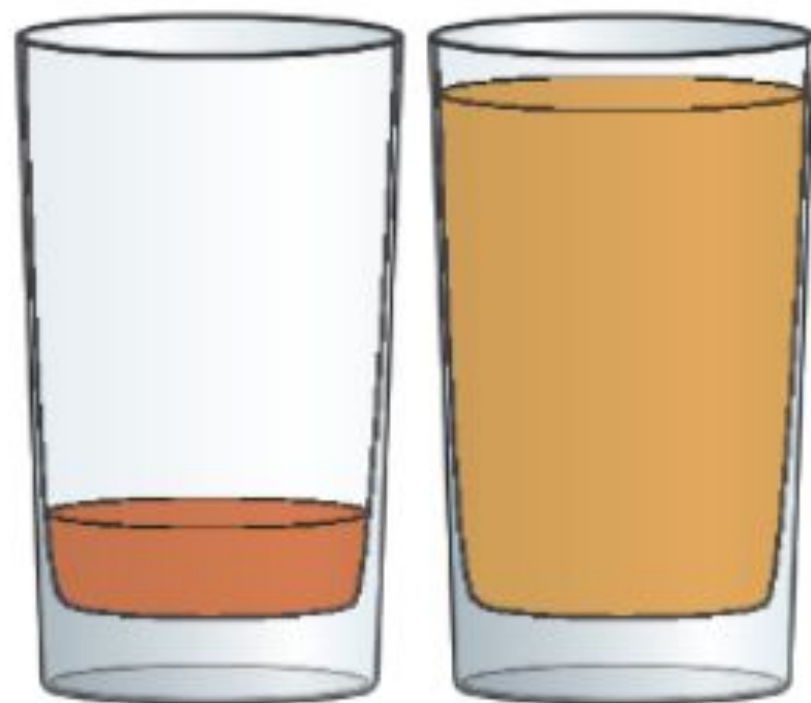
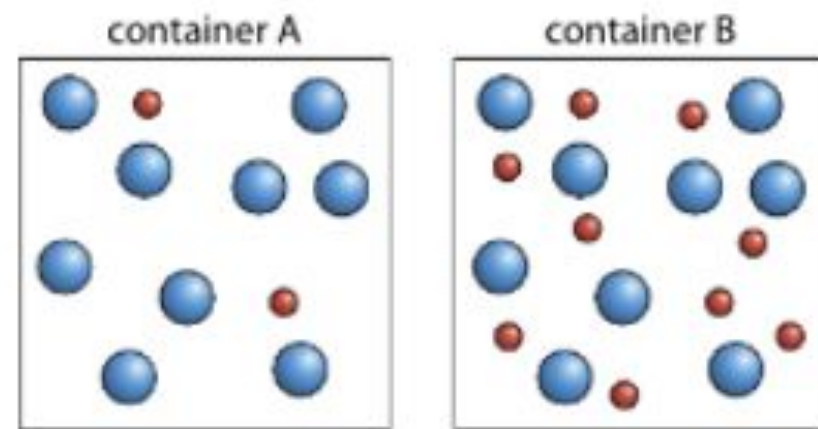
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Mark your work with a green pen!

Questions

- 1 **a** B
 b A
 - 2 Overall random movement of particles from an area of higher concentration to an area of lower concentration.
 - 3 **a** Gas; because the particles are not touching each other.
 b The concentrations are the same because there are the same number of blue particles in the same space/volume.
 - 4 **a** By diffusion; particles move randomly and spread from an area of high concentration to an area of low concentration.
 b Sofia is not correct; particles continue to move randomly even though the overall movement of the orange colour has stopped.
 - 5 A, C and D
 - 6 **c** The concentration is higher in B because there are more red particles in the same space than in A.
-

Let's practice...

Workbook page 64-68

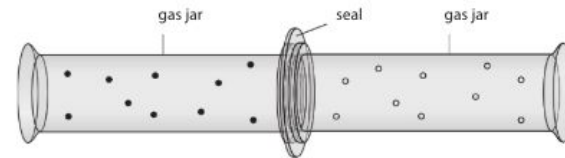
> 3.7 Particles on the move

Exercise 3.7A Diffusion in gases and liquids

Focus

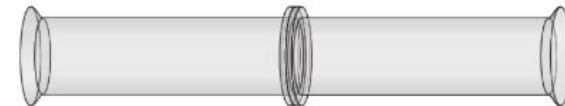
In this exercise, you will describe what happens when gases mix and when liquids mix.

- 1 Two glass jars are kept apart with a seal.
One jar contains a coloured gas.
The other jar contains a colourless gas.



The seal is removed.

Draw the arrangement of particles in the jars some time after the seal was removed.



Mark your work with a green pen

Topic 3.7 Particles on the move

Exercise 3.7A Diffusion in gases and liquids

- 1 Black circles and hollow circles randomly, and roughly evenly spread throughout the whole container; same number (10) of each type of particle.
- 2 Black circles randomly, and roughly evenly spread throughout the whole container; same number (10) of black circles.

Exercise 3.7B Diffusion

- 1 The random movement of particles from an area of higher concentration to an area of lower concentration.

- 2
 - a the particles/the ink diffuse(s); the particles start at high concentration and spread out randomly to areas of lower concentration
 - b diffusion occurs faster at higher temperature; because particles move faster at higher temperature
- 3 Close to the container the particles/brown gas is at high concentration; higher concentration means the particles are closer together/there are more particles in the space; the particles/brown gas diffuse; further away from the container, the particles are further apart.

Mark your work with a green pen

4 When the green colour fills the bottle, the movement of particles has stopped; **false**.

When the green colour fills the bottle, diffusion has stopped; **true**.

Exercise 3.7C Variables affecting diffusion

1 a The particles are closer together when the crystal first dissolves; the particles are in high concentration; the particles have not yet started to diffuse; particles randomly spread out to areas of lower concentration; when fully spread out, the particles are further apart.

Mark your work with a green pen

Exercise 3.7C Variables affecting diffusion

- 1 a** The particles are closer together when the crystal first dissolves; the particles are in high concentration; the particles have not yet started to diffuse; particles randomly spread out to areas of lower concentration; when fully spread out, the particles are further apart.
- b** Any two from: increasing the temperature; using more purple crystals; using a smaller volume of water.

Mark your work with a green pen

- 2 Diffusion happens faster at higher temperature; particles move faster at higher temperature; the particles from the hot coffee will change from being a liquid to being a gas faster/will evaporate faster; there will be a higher concentration of coffee particles in the air just above the hot coffee than the cold; particles from the hot coffee will take less time to travel from the cups to Zara's nose.
- 3 The concentration of particles causing the smell will be higher closer to the food; if the animal moves closer to the food, the smell will get stronger; if the animal moves away from the food, the smell will get less/weaker.

Extension task page 47-51 answers

Topic 3.3 Describing movement

Exercise 3.3A Distance/time graphs 1

1 the speed of an object

2 a C

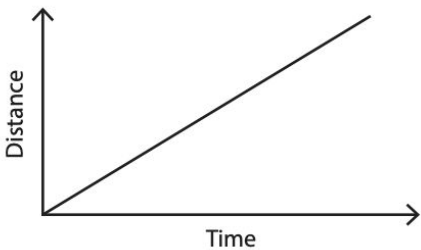
b B

Exercise 3.3B Distance/time graphs 2

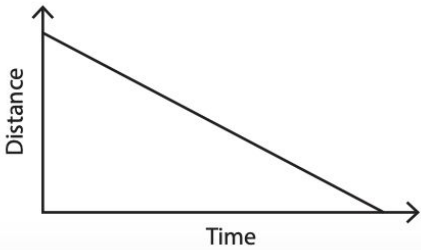
1 a



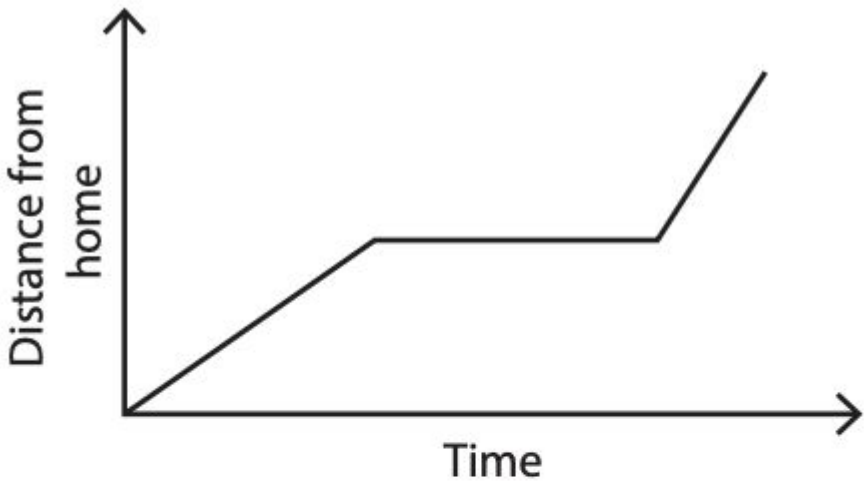
b



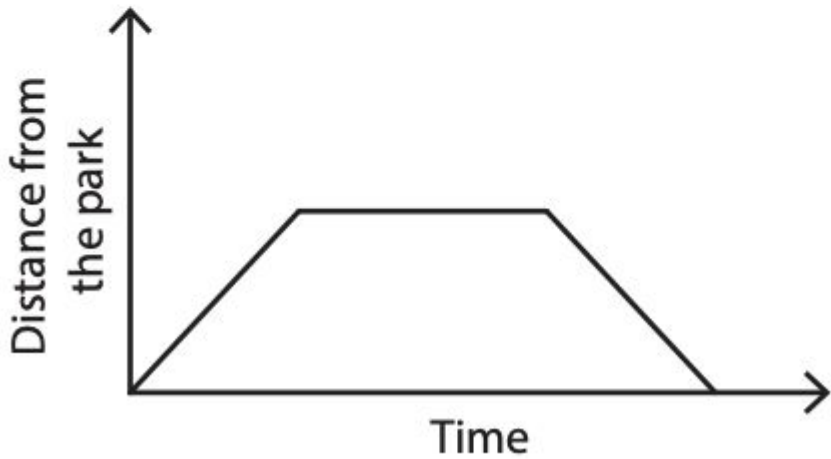
c



2



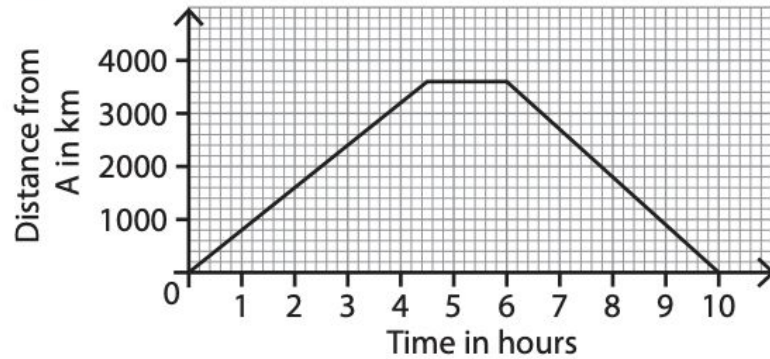
3



Extension task page 47-51 answers

Exercise 3.3C Distance/time graphs 3

1 a



b i 10 hours

i
$$\text{speed} = \frac{\text{distance}}{\text{time}}$$
$$= \frac{3600}{4.5}$$
$$= 800 \text{ km/h}$$

ii
$$\text{speed} = \frac{\text{distance}}{\text{time}}$$
$$= \frac{3600}{4}$$
$$= 900 \text{ km/h}$$

Topic 3.4 Turning forces

Exercise 3.4A Identifying turning forces

- 1 Ticks next to: pushing a door open, twisting the top off a bottle, and pushing the hands of a clock around.
- 2 Ticks next to the tap, gate and door handle.
- 3 moment